



Product Armor

Welcome

Brio Pharmaceuticals

Mr. GVL Prasad and Team

Controlled Variables, Specified Outcomes

Who are we?

Product Armor Packaging Pvt. Ltd.

 Founded in 2020 | Hyderabad, India



Cleanroom manufacturing company.



ISO Class 8 Certified Facility



Low Carbon Footprint company.



End-to-End Automation & Rigorous Quality Systems



Plans to expand into **drug delivery devices** and **advanced pharma products**



Offers a portfolio of HDPE bottles, Child Resistant Caps(CRC), and Continuous Thread (CT) Closures

Cleanroom Excellence

1,820 m²

ISO Class 8 Manufacturing Space

100%

Quality Control & Automated Inspection

What we do?

We are a cleanroom manufacturing company, leveraging advanced technology to serve the pharmaceutical industry with sustainable, low-carbon solutions.

Cleanroom Manufacturing

We specialize in cleanroom manufacturing, where minimal human contact ensures packaging that meets the strictest safety and contamination-free standards required for pharmaceutical products.

Pharmaceutical Industry

In the pharmaceutical industry, the reliability of containers is critical, any compromise in packaging can directly affect the efficacy of the medicine. Our cleanroom manufacturing setup is designed to meet these stringent requirements, making pharma a key sector we are uniquely positioned to serve.

Why Only HDPE Bottles & PP Caps?

We strategically started with HDPE bottles and PP caps, as they require minimal validation in the pharma industry. This allowed us to deliver trusted quality with faster compliance for our clients. Building on this strong foundation, we aim to become global partner of choice by manufacturing a wide range of cleanroom products.

Automated & Cost-Efficient Processes

- Fully automated production with minimal human intervention
- Consistent high quality, reduced errors
- Cost-efficient and scalable operations

Carbon Footprint & Sustainability

- Energy-efficient processes
- Lower carbon footprint
- Environmentally conscious approach

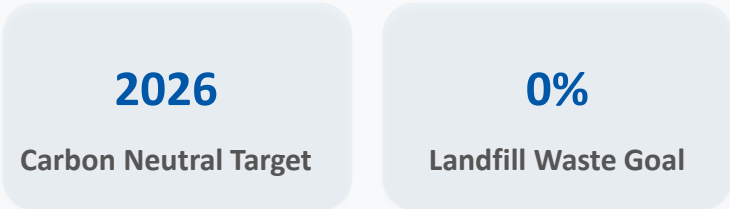
We operate in the **pharmaceutical packaging industry** where safety, compliance, and innovation are non-negotiable.

HDPE & PP in Pharma

HDPE (High-Density Polyethylene) bottles and PP (Poly Propylene) closures are among the most widely used primary packaging formats in the pharmaceutical industry.

- ✔ **Critical barrier** protecting medications from moisture & contamination
- ✔ Maintains **Drug Stability**
- ✔ **Light-resistant** properties protect light-sensitive medications

Product Armor's Sustainable Approach



Cleanroom Standards

Our ISO Class 8 cleanrooms maintain **strict particulate control** through HEPA filtration, ensuring product safety and regulatory compliance for global pharmaceutical markets.

Importance of Precision Manufacturing

Even minor variations in the product dimensions can compromise drug efficacy. Our **11-camera inspection system** ensures every bottle meets exact specifications, preventing potential medication degradation.

Sustainable Manufacturing Practices

Our low-carbon production processes reduce environmental impact while maintaining cleanroom manufacturing quality. We prioritize **energy-efficient** technologies throughout our manufacturing cycle.

HEPA : High-Efficiency Particulate Air (filter)

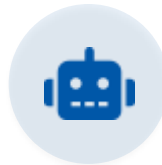
Our Vision & Mission

To become the **global partner of choice** by manufacturing pharmaceutical primary packaging and drug delivery devices of **unparallel quality**, through **sustainable and autonomous** manufacturing practices, without adding any extra cost to our customers.



Quality

Uncompromising standards with ISO Class 8
cleanroom manufacturing



Autonomous

Self-sufficient manufacturing with end-to-end
automation



Sustainability

Environmental responsibility with carbon
neutrality by 2026

Our Philosophy in Action

*When every variable is controlled, the outcome is **Guaranteed***

Controlled Variables

Material Procurement Standards

- Sourcing only certified, inspected, and validated raw materials
- Raw materials with clearly defined specifications under strict control

Process Precision

- Standardized manufacturing protocols
- Controlled temperature, pressure, and humidity
- Measurable benchmarks at every stage

Technology Advantage

- Automation systems controlling inputs
- Real-time monitoring & testing at each stage
- Minimal human intervention reducing variability

Design Intelligence

- Designed for cleanroom manufacturing & automation
- Optimized for minimal manpower and cost efficiency
- Functionality, safety, and sustainability embedded in design

Specified Outcomes

Optimal Product Quality

- Reduced variability, ensures reliability and quality in manufacturing.
- Contamination-free final products

Operational Predictability

- Reduced damage or delays
- Minimized risk of regulatory non-compliance and recalls
- Predictable, high-quality output for clients

Automation Impact

- Economies of Scale
- Efficient Resource Utilization
- Cost-Effective Production

Optimized by Design

- Robust, reliable and quality products
- Cost advantage & capability to manufacture any cleanroom product range
- Low carbon footprint

Child Resistant (CRC) and Continuous Thread (CT) Closures

What makes us different?

- Delivering Economies of Scale with a Low Carbon Footprint in Closures Manufacturing.
- Patented CRC manufacturing process that allows us to minimize our carbon footprint, delivering industry-leading sustainability.



Patent No. 568961 in CRC Manufacturing Technology

- **Designed with Sustainability in Mind:** All caps and closures are manufactured using eco-conscious, **low-carbon footprint** processes.
- **Automated** process with minimal human intervention
- **Rigorous inspection** and testing for **consistent quality**
- **Trusted Partner** of leading global Pharmaceutical brands
- **Cost-Efficient** Production Through **Economies of Scale**



Product Specifications



Sizes Available

28 mm

33 mm

38 mm

45 mm

53 mm



Production Scale

600M

CRC caps
annually

1B+

CT caps
annually



Quality Assurance



Cleanroom Compliant Manufacturing



100% visual inspection with multi-camera system



Automated checks for accuracy & dimensional consistency



Special Features



2D & 3D barcode printing as per client needs



CRC caps with **push-down & turn mechanism** for safety



Trademark pictorial + textual instructions on cap tops for ease of use



Low carbon footprint for sustainable manufacturing



Advanced HDPE bottles, manufactured in controlled Cleanroom environments to ensure contamination-free, pharma-grade quality.

- **Automated Production:** Minimal human intervention ensures precision and consistency.
- **Cleanroom Manufacturing:** Produced under controlled conditions for contamination-free, pharma-grade products.
- **Sustainable Manufacturing:** Energy-efficient processes minimize carbon footprint while maintaining top-quality standards.
- **Rigorous Visual Inspection:** Each bottle is inspected to guarantee defect-free packaging and high-quality standards.
- **Controlled Process Variables:** Temperature, pressure, and humidity are monitored to maintain uniform quality.
- **Consistent Product Specifications:** Oxygen barrier for medicine stability, durable construction for safe handling, and precise dimensional control for reliable fit, filling, and sealing.



30-2400

Size Range (cc)

227+ M

Annual Production



Sizes Available

30 cc

40 cc

60 cc

75 cc

100 cc

120 cc

150 cc

180 cc

200 cc

300 cc

400 cc

500 cc

625 cc

750 cc

850 cc

950 cc

1125 cc

1300 cc

1500 cc

2400 cc



Quality Assurance

- **Rigorous checks** with 11-camera inspection systems
- **100% visual inspection** for safety and compliance



Product Strengths

- **Low carbon footprint** for sustainable manufacturing
- **Automated Production** and low human intervention
- **Cleanroom Manufactured**
- **Variety of sizes** for diverse client needs



Applications

Designed for pharmaceutical products that demand cleanroom standards and rigorous contamination control.

Carbon-Efficient HDPE Bottles Manufacturing

ProductArmor

Our **CBF technology** reduces carbon emissions by up to **40%** compared to most competitors. This is achieved through lower operating temperatures, energy-efficient processes, and real-time automated controls.

Every ton of our products manufactured saves up to **0.8800 kg of CO2**, compared to standard industry methods.

Metric	Competitor Blow Moulding (Average)	Product Armor CBF Machine	Sources
Typical energy intensity	5–9.4 GJ/tonne polymer	3.5–6.5 GJ/tonne Polymer (est.)	Lawrence Berkeley National Laboratory Link 1
CO2 emissions <i>(Per every Tonne of Raw Material used in moulding)</i>	~1.8–2.2 tonnes of CO2 is emitted	~1.0–1.4 tonnes of CO2 is emitted <i>(30–40% lower)</i>	1.PT Online (Plastics Technology) 2.Nicerapid Link 1 , Link 2 , Link 3
Automation/SCADA optimization <i>(Supervisory Control And Data Acquisition)</i>	Limited	Extensive	India Pharma Outlook Link 1

GJ (Gigajoules): Unit of energy used in manufacturing. (1GJ = Equals 1 billion joules of energy, roughly the energy needed to power an average home for about a week.)
CO₂ (Carbon Dioxide): Greenhouse gas emitted during production.
Tonne: Standard unit (1,000 kg) used to measure materials or emissions.
GJ/tonne: A measure of energy consumed per tonne of material produced, used to compare manufacturing efficiency.

Our World-Class Facility

Our **cleanroom** manufacturing facility is built to deliver contamination-free, pharma-grade products. Equipped with **advanced automation**, energy-efficient with **low carbon footprint**, and sustainable technologies, it ensures consistent quality, operational excellence, and global compliance.



Our World-Class Facility

- **Stringent Cleanroom Policy:** Operates under ISO Class 8 standards with robust contamination control protocols ensuring **near-zero particulate presence**.
- **Minimal Human Intervention:** **Fully automated** systems and restricted access zones reduce contamination risk and maintain sterile conditions continuously.
- **Low Carbon Footprint:** **Energy-efficient** HVAC and filtration systems coupled with optimized process flows minimize power consumption and greenhouse gas emissions.
- **Zero Variables, Superior Practice:** Precision-controlled environment with consistent temperature, humidity, and airflow ensures flawless batch-to-batch reproducibility.
- **Flexible Manufacturing Capability:** Cleanroom optimized for diverse pharmaceutical packaging production, capable of handling complex projects with uncompromising quality.
- **Excellence Embedded:** Our facility embodies best-in-class practices that uphold product integrity, regulatory compliance, and environmental responsibility.

HVAC (Heating, Ventilation, and Air Conditioning): Controls temperature, humidity, and airflow to ensure stable, contamination-free conditions.



Our Key Competitive Advantages

01



Sustainable Practices

Carbon neutral commitment
with zero-waste
manufacturing

02



End-to-End Automation

Fully automated processes
from raw material to final
dispatch

03



Quality Approach

Zero compromise on quality
with rigorous inspection
systems

04



Economies of Scale

Strategic advantage
through high-capacity
infrastructure

05



Culture & Values

Work smart philosophy
with innovation-driven
culture

Manufacturing excellence meets environmental responsibility

Our sustainability roadmap focuses on **reducing environmental footprint** while maintaining the highest quality standards in pharmaceutical packaging.

CO₂

Carbon Neutral

- Committed to achieving carbon neutrality
- In-house renewable energy systems
- Carbon offset programs for unavoidable emissions



Water Conservation

- Rainwater harvesting systems
- Reduced dependency on external water sources



Zero-Waste Manufacturing

- Comprehensive recycling programs
- Composting for organic materials
- Minimized landfill waste



Fully automated, high-technology manufacturing

From raw materials to final dispatch, our **fully automated processes** ensure precision, scalability, and consistency with minimal human intervention.



Advanced Technology

Industrial Arms, Vision Systems, and Automated Special Purpose Machines



Quality Assurance

Designed to meet global cleanroom standards for pharmaceutical manufacturing.

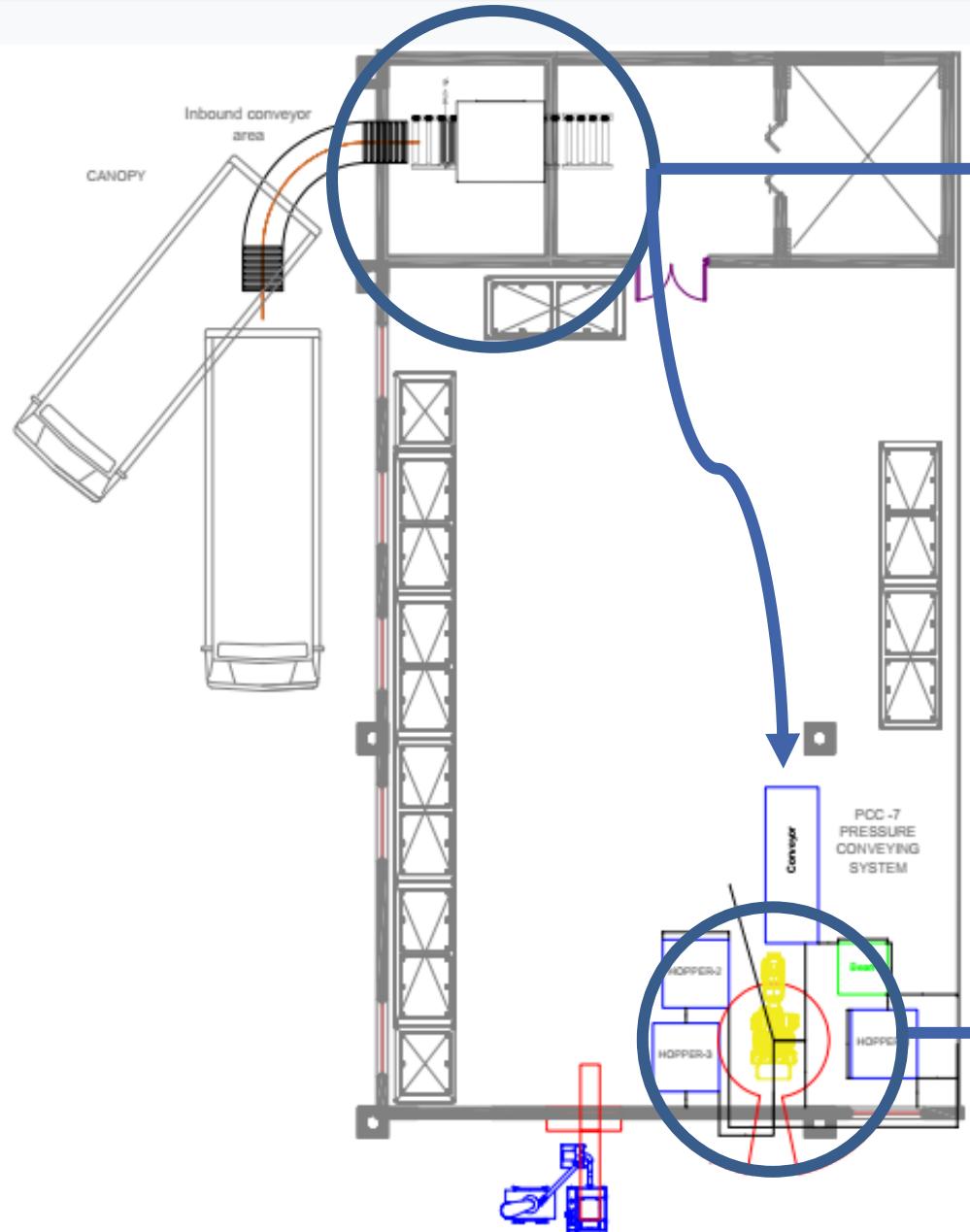


Future-Ready

Designed to scale and innovate for a rapidly evolving pharma industry.

From the very first step, our technology takes charge

Product
Armor



The First Step Towards Smart Manufacturing



Six Axis Arm Robot

Model FANUC, Imported from Japan • 1 unit installed



Capacity: ~120* bags per hour

Approx 25kgs each bag



Note : Values mentioned are per unit

Technologies in HDPE Bottles Production



Compression Blow Forming (CBF)

Model SACMI, Imported from Italy



Capacity

- Range : 100-200 CC
- Up to **120 bottles/min**
- Annual output: **63+ million** containers



Economies Of Scale

- High Output, Minimal Waste
- Automated, low Human Intervention
- Cost efficient, Delivering scalable solutions



Carbon Foot Print

- **30-35% reduction** in energy consumption
- **30°F (17°C) lower** melt temperatures



Sustainability

- Wastage rates **below 0.5%**
- Lowest in any polyolefin container process



30-35%

Energy Reduction

0.5%

Scrap Rate

1

Units Installed

Technologies in HDPE Bottles Production



Injection Blow Moulding (IBM)

Model JWM from China



Capacity

- Range : 30-1300 CC
- Up to **60 bottles/min**
- **Annual output: 31+ million** containers



Economies Of Scale

- 5 Units Installed, High Output
- **Automated**, low Human Intervention
- **Cost efficient**, Delivering scalable solutions



Carbon Foot Print

- **20% lower** energy usage per cycle
- Advanced process controls
- Highly efficient heating systems



Sustainability

- **Optimized** material usage
- **Minimized** waste production, energy-stable operations



20%

Energy Reduction

5

Units Installed

31M+

Annual Capacity

Technologies in HDPE Bottles Production



Extrusion Blow Moulding (EBM)

Model CMP, India



Capacity

- Range : 180-2400 CC
- Up to **6 bottles/min**
- **Annual output: 3+ million** containers



Economies Of Scale

- 2 Units Installed, High Output
- **Automated**, low Human Intervention
- **Cost efficient**, Delivering scalable solutions



Carbon Foot Print

- **15% reduced** electricity consumption



Sustainability

- Highly efficient plasticizing extruder
- Low wastage



15%

Energy Reduction

2

Units Installed

3+ M

Annual Capacity



Continuous Compression Moulding (CCM)

Model JOBO, imported from China



Capacity

- Range : 28-53 mm
- Up to **230 caps/min**
Annual output: 120+ million caps



Economies Of Scale

- **5 Units** Installed, High Output
- **Automated**, low Human Intervention
- **Cost efficient**, Delivering scalable solutions



Carbon Foot Print

- **Only technology in India** with patented compression molding for CRC caps with low carbon footprint
- **30% lower** energy consumption Versus traditional cap molding



Sustainability

- **Lowest verified** carbon footprint
- Industry-leading technology with **low material wastage**

30%

Energy Reduction

120+ M

Annual Capacity

5

Units Installed

0.9%

Scrap Rate



Injection Moulding (IM)

Model ZHAFIR, imported from China



Capacity

- **Range:** 45 mm
- Up to: **48 caps/min** (8 caps per 10 secs)
- **Annual Output:** ~100+ million caps



Economies Of Scale

- **1 Units** Installed, High Output
- **Automated**, less Human Intervention
- **Cost efficient**, Delivering scalable solutions



Carbon Foot Print

- Up to **65%** energy savings
- **Reduced CO2 emissions** with all-electric operation



Sustainability

- **Low carbon** footprint
- Low energy consumption



1

Units Installed

65%

Energy Reduction

100+ M

Annual Capacity

Wad Assembling Machines

Model Pentagon, India



Capacity

- **Annual Output:** ~120+ million Wads



Economies Of Scale

- **3 Units** Installed
- **Automated**
- **Cost efficient**, Delivering scalable solutions



Carbon Foot Print

- **Up to 20%** energy savings
- **Low carbon** footprint
- Efficient servo-driven systems



Precision

- Specially designed for wad assembly – Inner & Outer Caps
- **High accuracy** in cap placement
- **Auto roll feeding** & cutting systems



3

Units Installed

20%

Energy Reduction

120+ M

Annual Capacity

Delivering precision packaging solutions trusted by leading pharma brands worldwide



18+

Advanced machines



1B+

Units annual capacity



100%

Automated inspections

Powered by global technology and patented innovation

Customized Packaging Solutions

We go beyond standard packaging by offering **customizable solutions** to match every client's unique requirements.



Sizes & Capacities

- ✓ Wide range of closures
- ✓ Diverse bottle sizes
- ✓ Tailored to product-specific needs



Colors & Finishes

- ✓ Flexible branding options
- ✓ Multiple color choices
- ✓ Align with client identity



Design Variants

- ✓ Child Resistant Closures (CRC)
- ✓ Continuous Thread (CT) caps
- ✓ Diverse bottle designs



Thickness Options

- ✓ Optimized for strength
- ✓ Enhanced durability
- ✓ Cost-efficiency balanced



Special Features

- ✓ 1D/2D barcode printing
- ✓ Embossing options
- ✓ Brand-specific instructions



Work Smart, Not Hard

For us, working smart is not about doing less, it's about **achieving more** with



Focus

Prioritizing what drives the greatest impact and eliminating distractions.



Clarity

Setting clear goals and expectations so every effort is aligned with success.



Leveraging The Right Tools

Using technology, innovation, and resources to maximize productivity and deliver results effectively.

We believe **machines are built for repetition**, while **humans are built for innovation**.

Our culture is powered by automation, efficiency, and intelligence.



Automation First

If it can be automated, it will be



Speed with Precision

Delivering on time, with accuracy and reliability



Zero Compromise on Quality

Excellence is our non-negotiable baseline



Innovation in Every Role

Every Armorian is empowered to think like an entrepreneur

"At Product Armor, quality isn't optional; it's our standard."

Our Approach



Quality over revenue: No compromises, ever.



Prevention over correction: Systems built to eliminate errors before they occur.



Continuous improvement: Every batch, every process, every day.

Quality Systems in Place



11-camera vision system for 360° inspection of bottles



Single-camera system for closure inspection



Automated rejection system for defective units



Zero human bias technology ensures accuracy and consistency



Each batch is released through BMR Compliance



Monitoring of OOT for CTQ parameters

Our Certifications & Standards



ISO 9001:2015 **IN25/00000630**



ISO 15378:2017 **IN22/00000330**



USDMF **DMF 036675 DMF 037264**



Canadian DMF **MF2022-108**



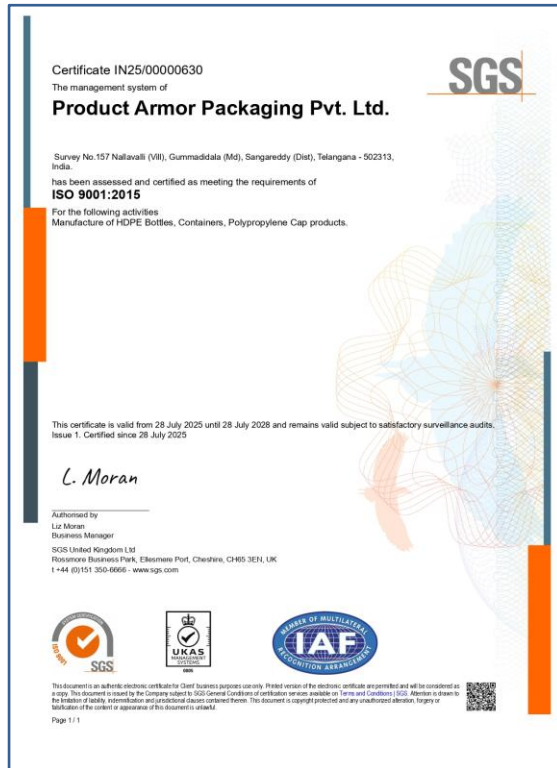
China DMF **B20220000820**

Our Certifications & Standards



ISO 9001:2015

Quality Management System that ensures every process is documented, monitored, and continually improved.



ISO 15378

Primary Packaging for Pharmaceuticals with certified compliance for cleanroom packaging materials, guaranteeing consumer safety and product integrity.



Regulatory Compliance

Our facilities and products meet stringent **US FDA, EU, Canadian and WHO** requirements for pharmaceutical packaging.

Continuous Validation ensures every line and process undergoes regular validation to ensure ongoing compliance and zero deviations.



100% Audit Preparedness



Zero Non-Conformance



Real-Time Documentation



Proactive Approach

Advanced Automation for Maximum Throughput

We have deliberately designed our cleanroom facility with advanced automation and integrated technologies to achieve maximum production scale without compromising quality or compliance.

Consistent Quality with Low Rejection Rates

Our fully automated, minimal human intervention manufacturing setup ensures consistently low rejection rates, driving down waste and costs significantly at higher volumes.

Scalable Production Capacity

The ability to produce billions of HDPE bottles and caps annually under tightly controlled cleanroom conditions delivers unmatched operational efficiency.

Energy-Efficient Operations for Cost Savings

Our facility's energy-optimized systems and low carbon footprint further reduce operational expenses, reinforcing cost competitiveness as production scales.

Competitive Advantage Through Scale & Technology

We have built true flexibility to scale seamlessly for any cleanroom manufacturing product, giving us capacity advantages over less agile competitors.

This scale combined with technology leadership translates to lower average costs per unit, enabling Product Armor to offer premium quality at highly competitive pricing.

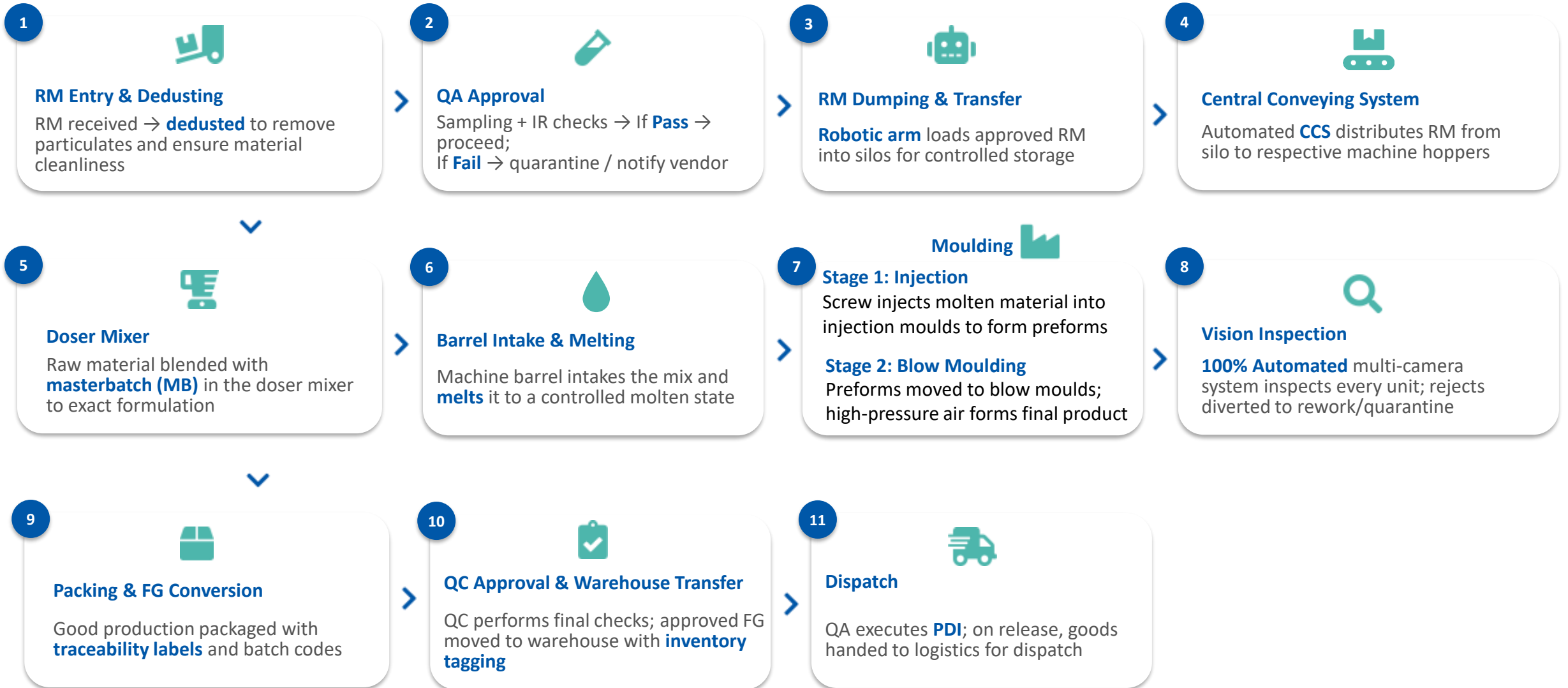
We deliver pharmaceutical primary packaging solutions, crafted under the highest cleanroom standards to ensure absolute protection of what's inside.



Our Promise

We protect what protects health. Every package we produce safeguards trust, care, and human well-being.

Manufacturing Process



Executive Steering Committee



Tapan Kumar
Plant Head



Syed Hussain
Quality Lead



Nimit Sharma
Head - Sales & BD



Deepika Vijaykumar
Head - Human Resources



Nilesh Shahji
Consultant - CFO

PRODUCT ARMOR





THANK YOU